

MAR ATHANASIOS COLLEGE FOR ADVANCED STUDIES, TIRUVALLA
(MACFAST)

(Affiliated to Mahatma Gandhi University, Kottayam)



MACFAST (AUTONOMOUS)

**MCACP107 SOFTWARE DEVELOPMENT LAB-I(C PROGRAMMING)
& C PROJECT**

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**MCACP107 SOFTWARE DEVELOPMENT LAB-I(C PROGRAMMING)
& C PROJECT**

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1.....

2.....

INDEX

SL.NO	DATE	NAME OF PROGRAM	PAGE NO
1	20/08/2025	Absolute value of a number using the conditional operator	7
2	20/08/2025	Check whether the given number is odd or even	9
3	20/08/2025	Calculate the simple interest.	11
4	25/08/2025	Swapping of 2 variables	13
5	29/08/2025	Check whether a year is leap year or not	15
6	29/08/2025	Calculate the area and perimeter of a circle	17
7	02/09/2025	Fahrenheit to Celsius Conversion	19
8	02/09/2025	Smallest among of three numbers using the nested if	21
9	08/09/2025	Arithmetic operations using switch.	23
10	08/09/2025	Decimal to base conversion using switch.	27
11	11/09/2025	Total and average of marks	35
12	11/09/2025	Factorial of a number	39
13	16/09/2025	Reverse of a number using while loop	41
14	16/09/2025	Sum of N natural numbers and their squares	43
15	16/09/2025	Occurrences of a digit in a number	45
16	19/09/2025	Check whether a number is palindrome or not	47
17	22/09/2025	Fibonacci series	49
18	22/09/2025	Check whether a number is Armstrong or not	51
19	25/09/2025	Prime numbers within a given range	53
20	25/09/2025	Root of a quadratic equation	55
21	03/10/2025	Multiplication table of a number using for loop	57
22	03/10/2025	Student grade calculation	59

SL.NO	DATE	NAME OF PROGRAM	PAGE NO
23	06/10/2025	Electricity bill calculation	63
24	06/10/2025	Sum of factorial series using functions	67
25	06/10/2025	Factorial of a number using recursion	69
26	13/10/2025	Display alphabet A to Z using loops	71
27	13/10/2025	Program to print the pattern	73
28	17/10/2025	Largest element in an array using functions	75
29	17/10/2025	Sum of two matrices using function	77
30	17/10/2025	Transpose of a matrix using function	79
31	22/10/2025	Linear search	81
32	24/10/2025	Binary search	83
33	03/11/2025	Bubble sort	87
34	07/11/2025	String handling operations	89
35	10/11/2025	Check whether a string is palindrome or not.	93
36	14/11/2025	String length without strlen()	95
37	14/11/2025	String copy without strcpy()	97
38	17/11/2025	String concatenation without strcat()	99
39	19/11/2025	String comparison without strcmp()	101
40	21/11/2025	Sorting of strings	103
41	21/11/2025	Call by value and call by reference	105
42	25/11/2025	Matrix multiplication using pointers	107
43	28/11/2025	Vowel count in a string	111
44	01/12/2025	Personal data using structures	113

SL.NO	DATE	NAME OF PROGRAM	PAGE NO
45	04/12/2025	File read and display	117
46	08/12/2025	File copy using command line arguments	119
47	10/12/2025	Count words and lines in a file	123
48	12/12/2025	Count character types in a file	127
49	16/12/2025	Employee file creation and display	129
50	18/12/2025	Mismatch of brackets in a file	133
51	13/12/2025	Project – BANKING MANAGEMENT SYSTEM	137

OUTPUT

```
Enter a number  
-19  
19 is an absolute value_
```

```
Enter a number  
19  
19 is an absolute value_
```

PROGRAM – 1

AIM : Program to find the absolute value using conditional operator.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int num, absvalue;
    clrscr();
    printf("Enter a number\n");
    scanf("%d",&num);
    absvalue = (num < 0)? -num:num;
    printf("%d is an absolute value",absvalue);

    getch();
}
```

OUTPUT

```
Enter a Number  
1  
1 is an odd number
```

```
Enter a Number  
2  
2 is an even number
```


PROGRAM – 2

AIM : Program to check whether the given number is odd or even.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a;
    clrscr();
    printf("Enter a Number\n");
    scanf("%d",&a);

    if(a%2==0)
    {
        printf("%d is an even number",a);
    }
    else
    {
        printf("%d is an odd number",a);
    }
    getch();
}
```

OUTPUT

```
Enter Your initail Amount:  
500  
Enter Rate of interest:  
2.5  
Enter Time (in years):  
5  
Your interest: 62.500000
```

PROGRAM – 3

AIM : Program to find simple interest.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float P,T,SI,R;
    clrscr();
    printf("Enter Your initail Amount:\n");
    scanf("%f",&P);
    printf("Enter Rate of interest:\n");
    scanf("%f",&R);
    printf("Enter Time (in years):\n");
    scanf("%f",&T);
    SI =(P*R*T)/100;
    printf("Your interest: %f",SI);
    getch();
}
```

OUTPUT

```
Enter the first no:  
111  
Enter the second no:  
333  
Before Swapping: a= 111, b= 333  
After Swapping: a= 333,b= 111
```

PROGRAM – 4

AIM : Program to interchange the value of 2 variables (swapping).

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,temp=0;
    clrscr();
    printf("Enter the first no:\n");
    scanf("%d",&a);
    printf("Enter the second no: \n");
    scanf("%d",&b);
    printf("Before Swapping: a= %d, b= %d\n",a,b);
    temp =a;
    a=b;
    b=temp;
    printf("After Swapping: a= %d,b= %d",a,b);
    getch();
}
```

OUTPUT

```
Enter a year:  
2010  
2010 is not a leap year_
```

```
Enter a year:  
2020  
2020 is a leap year._
```

PROGRAM – 5

AIM : Program to check whether a year is a leap year or not.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int year;
    clrscr();
    printf("Enter a year:\n");
    scanf("%d",&year);
    if((year % 4 == 0) || (year % 400 == 0))
    {
        printf("%d is a leap year.",year);
    }
    else{
        printf("%d is not a leap year");
    }
    getch();
}
```

OUTPUT

```
Enter the radius of the Circle:  
5  
Area of the Circle is: 78.500000  
Perimeter of Circle is: 31.400000  
—
```


PROGRAM – 6

AIM : Program to calculate the area and perimeter of a circle.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#define PI 3.14
void main()
{
    float r,perimeter,area;
    clrscr();
    printf("Enter the radius of the Circle:\n");
    scanf("%f",&r);
    area = PI * r * r;
    perimeter = 2 * PI *r;
    printf("Area of the Circle is: %f\n",area);
    printf("Perimeter of Circle is: %f\n",perimeter);
    getch();
}
```

OUTPUT

```
Enter Temperature in Fahrenheit:  
45  
Temperature in Celcius: 7.222222_
```

PROGRAM – 7

AIM : Program to accept Temperature in Fahrenheit and convert it to degree celcius.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    float f,c;
    clrscr();
    printf("Enter Temperature in Fahrenheit:\n");
    scanf("%f",&f);
    c = ((f - 32) * 5.0) / 9.0;
    printf("Temperature in Celcius: %f",c);
    getch();
}
```

OUTPUT

```
Enter 1st number:  
45  
Enter 2nd number:  
33  
Enter 3rd number:  
91  
33 is the lowest number.
```

PROGRAM – 8

AIM : Program to find the lowest of 3 numbers using nested if statement.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int low, first, second, third;
    clrscr();
    printf("Enter 1st number:\n");
    scanf("%d", &first);
    printf("Enter 2nd number:\n");
    scanf("%d", &second);
    printf("Enter 3rd number:\n");
    scanf("%d", &third);
    if (first < second)
    {
        if (first < third)
            low = first;
        else
            low = third;
    }
    else
    {
        if (second < third)
            low = second;
        else
            low = third;
    }
    printf("%d is the lowest number.", low);
    getch();
}
```

OUTPUT

Choose An Operator

1. Addition (+)
2. Subtraction (-)
3. Multiply (*)
4. Division (/)
5. Modulus (%)
6. EXIT

Enter a choice:

1

Enter 1st number

10

Enter 2nd number

5

Addition

10 + 5 = 15

Enter a choice:

2

Enter 1st number

10

Enter 2nd number

5

Subtraction

10 - 5 = 5

PROGRAM – 9

AIM : Program to implement arithmetic operation using switch.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
    int choice;
    int a,b,sum,product,subtract;
    float div,mod;
    clrscr();
    while(1)
    {
        printf("\n");
        printf("Choose An Operator \n\n");
        printf("1. Addition ( + )\n");
        printf("2. Substraction ( - )\n");
        printf("3. Multiply ( * )\n");
        printf("4. Division ( / )\n");
        printf("5. Modulus ( % )\n");
        printf("6. EXIT \n");
        printf("Enter a choice: \n");
        scanf("%d",&choice);
        if (choice == 6){
            exit(0);
        }
        printf("Enter 1st number\n");
        scanf("%d",&a);
        printf("Enter 2nd number\n");
        scanf("%d",&b);
        switch(choice){
```

OUTPUT

```
Enter a choice:
3
Enter 1st number
10
Enter 2nd number
5
Multiply
10 * 5 = 50
```

```
Enter a choice:
4
Enter 1st number
10
Enter 2nd number
5
Division
10 / 5 = 2.000000
```

```
Enter a choice:
5
Enter 1st number
10
Enter 2nd number
5
Modulus
10 % 5 = 0.000000
```


case 1:

```
printf("Addition\n");
sum = a + b;
printf("%d + %d = %d",a,b,sum);
break;
```

case 2:

```
printf("Substraction\n");
subtract = a - b;
printf("%d - %d = %d",a,b,subtract);
break;
```

case 3:

```
printf("Multiply\n");
product = a * b;
printf("%d * %d = %d",a,b,product);
break;
```

case 4:

```
printf("Division\n");
div = a / b;
printf("%d / %d = %f",a,b,div);
break;
```

case 5:

```
printf("Modulus\n");
mod = a % b;
printf("%d %% %d = %f",a,b,mod);
break;
```

default:

```
printf("Invalid Input");
```

```
}
```

```
}
```

```
getch();
```

```
}
```

OUTPUT

```
Choose conversion
1. Binary
2. Octal
3. Hexadecimal
4. Exit
Enter choice: 1
Enter a decimal no: 15
Binary: 1111
```

```
Choose conversion
1. Binary
2. Octal
3. Hexadecimal
4. Exit
Enter choice: 2
Enter a decimal no: 15
Octal: 17
```

```
Choose conversion
1. Binary
2. Octal
3. Hexadecimal
4. Exit
Enter choice: 3
Enter a decimal no: 15
Hexadecimal: F
```

PROGRAM – 10

AIM : Program to convert decimal numbers to different bases using switch.

SOURCE CODE:

```
#include<stdio.h>
#include<stdlib.h>
void contobin(int num);
void contoact(int num);
void contohehexa(int num);
int main()
{
    int num, choice;
    clrscr();
    while (1)
    {
        printf("\nChoose conversion\n");
        printf("1. Binary\n");
        printf("2. Octal\n");
        printf("3. Hexadecimal\n");
        printf("4. Exit\n");
        printf("Enter choice:\t");
        scanf("%d", &choice);
        if(choice != 4){
            printf("\nEnter a decimal no:\t");
            scanf("%d",&num);
        }
        switch (choice)
        {
            case 1:
                contobin(num);
                break;
```


case 2:

 contooct(num);

 break;

case 3:

 contohehexa(num);

 break;

case 4:

 exit(0);

default:

 printf("Invalid choice\n");

 }

 printf("\n");

}

return 0;

}

void contobin(int num)

{

 int bin[32], i = 0, j;

 if (num == 0)

 {

 printf("Binary:\t0\n");

 return;

 }

 while (num > 0)

 {

 bin[i++] = num % 2;

 num /= 2;

 }


```

printf("Binary:\t");
for (j = i - 1; j >= 0; j--)
    printf("%d", bin[j]);
}

```

```

void contoocot(int num)
{
    int oct[32], i = 0, j;
    if (num == 0)
    {
        printf("Octal:\t0\n");
        return;
    }
    while (num > 0)
    {
        oct[i++] = num % 8;
        num /= 8;
    }
    printf("Octal:\t");
    for (j = i - 1; j >= 0; j--)
        printf("%d", oct[j]);
}

```

```

void contohehexa(int num)
{
    char hex[32];
    int i = 0, j, rem;
    if (num == 0)
    {
        printf("Hexadecimal:\t0\n");
        return;
    }
    while (num > 0)
    {

```



```
rem = num % 16;
    hex[i++] = (rem < 10) ? rem + '0' : (rem - 10) + 'A';
    num /= 16;
}
printf("Hexadecimal:\t");
for (j = i - 1; j >= 0; j--)
    printf("%c", hex[j]);
}
```

OUTPUT

```
=== ENTER STUDENT'S MARKS ===
```

```
Enter Mark of Subject1 (outof 100):  
55
```

```
Enter Mark of Subject2 (outof 100):  
91
```

```
Enter Mark of Subject3 (outof 100):  
79
```

```
Enter Mark of Subject4 (outof 100):  
63
```

```
Enter Mark of Subject5 (outof 100):  
81
```

```
===== STUDENT'S RESULT =====
```

```
Sub ject1:      55
```

```
Sub ject2:      91
```

```
Sub ject3:      79
```

```
Sub ject4:      63
```

```
Sub ject5:      81
```

```
Total Marks (Outof 500) : 369
```

```
Average : 73.800000
```

PROGRAM – 11

AIM : Program to find total and avrage of 5 marks of student.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int sum;
    char name[20];
    float avg;
    int mark1,mark2,mark3,mark4,mark5;
    clrscr();
    printf("=== ENTER STUDENT'S MARKS ===\n\n");
    printf("Enter Mark of Subject1 (outof 100):\n");
    scanf("%d",&mark1);
    printf("Enter Mark of Subject2 (outof 100): \n");
    scanf("%d",&mark2);
    printf("Enter Mark of Subject3 (outof 100): \n");
    scanf("%d",&mark3);
    printf("Enter Mark of Subject4 (outof 100): \n");
    scanf("%d",&mark4);
    printf("Enter Mark of Subject5 (outof 100): \n");
    scanf("%d",&mark5);
    sum = mark1+mark2+mark3+mark4+mark5;
    avg = (mark1+mark2+mark3+mark4+mark5)/5;
    printf("\n");
    printf("===== STUDENT'S RESULT =====\n");
    printf("Subject1:\t %d\n",mark1);
    printf("Subject2:\t %d\n",mark2);
    printf("Subject3:\t %d\n",mark3);
    printf("Subject4:\t %d\n",mark4);
```



```
printf("Subject5:\t %d\n",mark5);  
printf("\n");  
printf("\n Total Marks (Outof 500) : %d",sum);  
printf("\n Average : %f",avg);  
getch();  
}
```

OUTPUT

```
Enter a number to find factorial: 5  
Factorial of 5 is 120
```

```
Enter a number to find factorial: 3  
Factorial of 3 is 6
```

PROGRAM – 12

AIM : Program to find factorial of a number.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i,n,f=1;
    clrscr();
    printf("Enter a number to find factorial: ");
    scanf("%d",&n);

    for(i=1;i<=n;i++)
    {
        f = f * i;
    }
    printf("Factorial of %d is %d",n,f);
    getch();
}
```

OUTPUT

```
Enter a number: 916  
Reversed number = 619
```

```
Enter a number: 311  
Reversed number = 113
```

```
—
```


PROGRAM – 13

AIM : Program to find reverse of a number using while loop.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int num, reversed = 0, remainder;
    printf("Enter a number: ");
    scanf("%d", &num);
    while (num != 0)
    {
        remainder = num % 10;
        reversed = reversed * 10 + remainder;
        num = num / 10;
    }
    printf("Reversed number = %d\n", reversed);
    getch();
}
```

OUTPUT

```
Enter the value of N: 5
Sum of first 5 natural numbers = 15
Sum of squares of first 5 natural numbers = 55
_
```

```
Enter the value of N: 3
Sum of first 3 natural numbers = 6
Sum of squares of first 3 natural numbers = 14
_
```

PROGRAM – 14

AIM : Program to generate sum of N natural numbers and their squares

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main() {
    int n, i;
    int sum = 0, squareSum = 0;
    clrscr();
    printf("Enter the value of N: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        sum += i;
        squareSum += i * i;
    }
    printf("Sum of first %d natural numbers = %d\n", n, sum);
    printf("Sum of squares of first %d natural numbers = %d\n", n, squareSum);
    getch();
}
```

OUTPUT

```
Enter a number: 1771
Enter the digit to count: 7
The digit 7 occurs 2 times.
-
```

```
Enter a number: 8881
Enter the digit to count: 8
The digit 8 occurs 3 times.
-
```

PROGRAM – 15

AIM : Program to find the occurrences of a digit in a number

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main() {
    int num, digit, count = 0, remainder;
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &num);
    printf("Enter the digit to count: ");
    scanf("%d", &digit);
    while (num != 0) {
        remainder = num % 10;
        if (remainder == digit)
        {
            count++;
        }
        num = num / 10;
    }
    printf("The digit %d occurs %d times.\n", digit, count);
    getch();
}
```

OUTPUT

```
Enter a number: 121  
Its a palindrome number
```

```
Enter a number: 331  
Its not palindrome number
```

PROGRAM – 16

AIM : Program to check whether a number is palindrome or not.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int num,rev=0,rem,temp;
    clrscr();
    printf("Enter a number: ");
    scanf("%d",&num);
    temp = num;
    while(num > 0)
    {
        rem = num % 10;
        rev = rev * 10 + rem;
        num = num /10;
    }
    if(temp == rev)
        printf("palindrome",temp);
    else
        printf(" not palindrome \n",temp);
    getch();
}
```

OUTPUT

```
Enter the limit:  
10  
0 1 1 2 3 5 8 13 21 34 _
```

```
Enter the limit:  
7  
0 1 1 2 3 5 8 _
```


PROGRAM – 17

AIM : Program to print Fibonacci series up to a particular number.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,i,nextTerm;
    int a = 0, b = 1;
    printf("Enter the limit: \n");
    scanf("%d",&n);
    printf("%d %d ",a,b);
    for(i = 2; i < n; i++){
        nextTerm = a + b;
        printf("%d ",nextTerm);
        a = b;
        b = nextTerm;
    }
    getch();
}
```

OUTPUT

```
Enter a number :      153
Enterd number is armstrong number
```

```
Enter a number :      132
Entered number is not armstrong
```

PROGRAM – 18

AIM : Program to check whether a number is Armstrong or not.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int num,rem,arm=0,cpy;
    clrscr();
    printf("Enter a number : \t");
    scanf("%d",&num);
    cpy=num;
    while(num>0)
    {
        rem=num%10;
        arm=arm+(rem*rem*rem);
        num=num/10;
    }
    if(arm==cpy)
        printf("Entered number is armstrong number");
    else
        printf("Entered number is not armstrong");
    getch();
}
```

OUTPUT

```
Enter the starting number: 2
```

```
Enter the ending number: 40
```

```
Prime numbers between 2 and 40 are:
```

```
2 3 5 7 11 13 17 19 23 29 31 37 _
```

PROGRAM – 19

AIM : Program to print prime numbers within a given range.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int start, end, i, j, flag;
    clrscr();
    printf("Enter the starting number: ");
    scanf("%d", &start);
    printf("Enter the ending number: ");
    scanf("%d", &end);
    printf("\nPrime numbers between %d and %d are:\n", start, end);
    for(i = start; i <= end; i++)
    {
        if(i < 2)
            continue;
        flag = 0;
        for(j = 2; j <= i/2; j++)
        {
            if(i % j == 0)
            {
                flag = 1;
                break;
            }
        }
        if(flag == 0)
            printf("%d ", i);
    }
    getch();
}
```

OUTPUT

```
Enter coefficients a,b and c:  
1  
5  
6  
Roots are real and different:  
Root 1 = -2.00  
Root 2 = -3.00  
_
```

```
Enter coefficients a,b and c:  
12  
11  
9  
No real roots_
```

PROGRAM – 20

AIM : Program to find root of a quadratic equation.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
    float a,b,c,discri,root1,root2;
    clrscr();
    printf("Enter coefficents a,b and c: \n");
    scanf("%f %f %f",&a,&b,&c);
    discri = (b * b) - (4 * a * c);
    if(discri >= 0)
    {
        root1 = ( -b + sqrt(discri)) / (2 * a);
        root2 = ( -b - sqrt(discri)) / (2 * a);
        printf("Roots are real and different: \n");
        printf("Root 1 = %.2f\n",root1);
        printf("Root 2 = %.2f\n",root2);
    }
    else{
        printf("No real roots");
    }
    getch();
}
```

OUTPUT

```
Enter a number to show its multiplication table: 9
Enter the end point of the multiplication table: 15
Multiplication Table of 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
9 * 11 = 99
9 * 12 = 108
9 * 13 = 117
9 * 14 = 126
9 * 15 = 135
```


PROGRAM – 21

AIM : Program to print multiplication table of a number using for loop.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int num,limit,i,m;
    clrscr();
    printf("Enter a number to show its multiplication table.");
    scanf("%d",&num);
    printf("Enter the end point of the multiplication table.");
    scanf("%d",&limit);
    printf("Multiplication Table of %d",num);
    for(i=1;i<=limit;i++)
    {
        m = num * i;
        printf("%d * %d = %d",num,i,m);
    }
    getch();
}
```

OUTPUT

```
Enter the marks of the student:  
90  
Student Received : A Grade_
```

```
Enter the marks of the student:  
45  
Student Received : E Grade
```

```
Enter the marks of the student:  
28  
Student Failed in the examination
```

PROGRAM – 22

AIM: Program to read marks of a student and find their grade according to the mark using else...if ladder.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int marks;
    clrscr();
    printf("Enter the marks of the student:\n");
    scanf("%d",&marks);
    if(marks <= 100)
    {
        if(marks >= 90 && marks <= 100)
        {
            printf("Student Received : A Grade");
        }
        else if(marks >= 80 && marks < 90)
        {
            printf("Student Received : B Grade");
        }
        else if(marks >= 70 && marks < 80)
        {
            printf("Student Received : C Grade");
        }
        else if(marks >= 60 && marks < 70)
        {
            printf("Student Received : D Grade");
        }
        else if(marks >= 35 && marks < 60)
        {
```



```
printf("Student Received : E Grade");  
}  
else  
{  
printf("Student Failed in the examination \n");  
}  
}  
else{  
printf("Marks is outof 100.\n Please enter marks within 100");  
}  
  
getch();  
}
```

OUTPUT

```
Enter the customer id: 121
Enter the Power Consumed: 563

===== Your Bill =====

Customer ID: 121
Power Consumed By Customer: 563
Total Amount to Pay is 680.400024
```

PROGRAM – 23

AIM : An electric power distribution company charges its domestic consumer as follows:

CONSUMPTION UNITS	RATE OF CHARGE
0 - 200	Rs: 0.50 per unit
201 - 400	Rs: 100 plus Rs: 0.65 per unit excess of 200
401 - 600	Rs: 230 plus Rs: 0.80 per unit excess of 400
601 and above	Rs: 390 plus Rs: 1.00 per unit excess of 600

Read the customer number and power consumed and prints the amount to be paid by the customer.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int cust_no,units;
    float amount;
    clrscr();
    printf("Enter the customer id: \t");
    scanf("%d",&cust_no);
    printf("Enter the Power Consumed: \t");
    scanf("%d",&units);
    if(units > 0 && units <= 200)
    {
        amount = units * 0.50;
    }
    else if(units >= 201 && units <= 400)
    {
        amount = (units * 0.65) + 100;
    }
    else if(units >=401 && units <= 600)
```



```
{  
    amount = (units * 0.80) + 230;  
}  
else if(units >=601)  
{  
    amount = (units * 1.00) + 390;  
}  
else  
{  
    printf("Invalid Input ");  
}  
printf("\n");  
printf("===== Your Bill =====");  
printf("\n");  
printf("\nCustomer ID: %d",cust_no);  
printf("\nPower Consumed By Customer: %d",units);  
printf("\nTotal Amount to Pay is %f",amount);  
    getch();  
}
```

OUTPUT

```
Enter the value of N:    3
Sum of series 1! + 2! + ... + 3! = 9
```

```
Enter the value of N:    5
Sum of series 1! + 2! + ... + 5! = 153
```

```
Enter the value of N:    10
Sum of series 1! + 2! + ... + 10! = 4037913
```

PROGRAM – 24

AIM : Program to find sum of series $1! + 2! + 3! + \dots + N!$ Using Function.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
long factorial(int n) {
    long long fact = 1;
    int i;
    for (i = 1; i <= n; i++) {
        fact = fact * i;
    }
    return fact;
}
long sumOfSeries(int N) {
    long sum = 0;
    int i;
    for (i = 1; i <= N; i++) {
        sum += factorial(i);
    }
    return sum;
}
void main() {
    int N;
    long result;
    clrscr();
    printf("\n");
    printf("Enter the value of N:\t ");
    scanf("%d", &N);
    result = sumOfSeries(N);
    printf("Sum of series  $1! + 2! + \dots + %d! = %lld$ \n", N, result);
    getch();
}
```

OUTPUT

```
Enter a number: 3
Factorial of 3 = 6
-
```

```
Enter a number: 4
Factorial of 4 = 24
-
```

```
Enter a number: 5
Factorial of 5 = 120
```

PROGRAM – 25

AIM : Program to find factorial of a number using recursion.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
long factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    return n * factorial(n - 1);
}
void main() {
    int num;
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &num);
    if (num < 0) {
        printf("Factorial of negative numbers is not defined.\n");
    } else {
        printf("Factorial of %d = %lld\n", num, factorial(num));
    }
    getch();
}
```

OUTPUT

```
===== Alphabet Order From A to Z =====
```

```
A      B      C      D      E      F      G      H      I      J  
K      L      M      N      O      P      Q      R      S      T  
U      V      W      X      Y      Z
```

PROGRAM – 26

AIM : Program to display alphabet A to Z using loops.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    char i;
    clrscr();
    printf("===== Alphabet Order From A to Z =====\n");
    for(i='A';i<='Z';i++)
    {
        printf("%c\t",i);
    }
    getch();
}
```

OUTPUT

```
Enter the limit: 5
```

```
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * * *
```


PROGRAM – 27

AIM : Program to print the pattern.

```
*
* *
* * *
* * * *
```

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i,j,n;
    clrscr();
    printf("Enter the limit:\t");
    scanf("%d",&n);
    printf("\n");
    for(i=0;i<=n;i++)
    {
        for(j=0;j<=i;j++)
        {
            printf("* ");
        }
        printf("\n");
    }
    getch();
}
```

OUTPUT

```
Enter the number of elements: 5
Enter 5 numbers:
91
101
22
19
110
The largest number is: 110
```

PROGRAM – 28

AIM : Program to read an array of n numbers; find the largest among them and display using functions.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
int findLargest(int arr[], int n) {
    int i, max = arr[0];
    for (i = 1; i < n; i++) {
        if (arr[i] > max) {
            max = arr[i];
        }
    }
    return max;
}
void main() {
    int n, i;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d numbers:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    int largest = findLargest(arr, n);
    printf("The largest number is: %d\n", largest);
    getch();
}
```

OUTPUT

```
Enter the number of rows and columns of matrix: 2 2
Enter the elements of first matrix:
4
5
6
7
Enter the elements of second matrix:
1
0
4
3
Addition of two matrices:
5      5
10     10
```

PROGRAM – 29

AIM: Program to find the sum of two matrices using function.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void addition(int r,int c,int a[10][10],int b[10][10]);
void main() {
    int r, c, a[10][10], b[10][10],i, j;
    printf("Enter the number of rows and columns of matrix: ");
    scanf("%d %d",&r,&c);
    printf("Enter the elements of first matrix:\n");
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
    printf("Enter the elements of second matrix:\n");
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &b[i][j]);
    addition(r,c,a,b);
    getch();
}
void addition(int r,int c,int a[10][10],int b[10][10])
{ int add[10][10],i,j;
  for (i = 0; i < r; i++)
      for (j = 0; j < c; j++)
          add[i][j] = a[i][j] + b[i][j];
  printf("Addition of two matrices: \n");
  for (i = 0; i < r; i++){
      for (j = 0; j < c; j++)
          printf("%d\t", add[i][j]);
      printf("\n");
  }
}
```

OUTPUT

```
Enter number of rows: 2
Enter number of columns: 3
Enter matrix elements:
1
2
3
4
5
6
Transpose of the matrix:
1 4
2 5
3 6
-
```

```
Enter number of rows: 2
Enter number of columns: 2
Enter matrix elements:
1
2
3
4
Transpose of the matrix:
1 3
2 4
```

PROGRAM – 30

AIM: Program to find the transpose of a matrix using function.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void transpose(int a[10][10], int r, int c) {
    int t[10][10], i, j;
    for(i = 0; i < r; i++) {
        for(j = 0; j < c; j++) {
            t[j][i] = a[i][j];
        }
    }
    printf("Transpose of the matrix:\n");
    for(i = 0; i < c; i++) {
        for(j = 0; j < r; j++) {
            printf("%d ", t[i][j]);
        }
        printf("\n");
    }
}
void main() {
    int a[10][10], r, c, i, j;
    printf("Enter number of rows: \t");
    scanf("%d", &r);
    printf("Enter number of columns: \t");
    scanf("%d", &c);
    printf("Enter matrix elements:\n");
    for(i = 0; i < r; i++) {
        for(j = 0; j < c; j++) {
            scanf("%d", &a[i][j]);
        }
    }
    transpose(a, r, c);
    getch();
}
```

OUTPUT

```
Enter number of elements: 5
Enter 5 elements:
10
91
77
11
0
Enter element to search: 77
Element 77 found at position 3
```

```
Enter number of elements: 4
Enter 4 elements:
9
11
20
67
Enter element to search: 5
Element 5 not found in the array_
```


PROGRAM – 31

AIM: Program to search an element in an array using linear search.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a[50], n, key, i, found = 0;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    printf("Enter element to search: ");
    scanf("%d", &key);
    for (i = 0; i < n; i++)
    {
        if (a[i] == key)
        {
            printf("Element %d found at position %d", key, i + 1);
            found = 1;
            break;
        }
    }
    if (found == 0)
        printf("Element %d not found in the array", key);
    getch();
}
```

OUTPUT

```
Enter number of elements: 5
Enter 5 elements in ascending order:
11
23
34
76
87
Enter element to search:      19
Element not found_
```

```
Enter number of elements: 5
Enter 5 elements in ascending order:
11
21
41
51
91
Enter element to search:      51
Element found at position 4
```

PROGRAM – 32

AIM: Program to search an element in an array using binary search.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a[20],n,i,key;
    int low = 0,high,mid,found = 0;
    clrscr();
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter %d elements in ascending order: \n",n);
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter element to search: \t");
    scanf("%d",&key);
    high = n - 1;
    while(low <= high)
    {
        mid = (low + high)/2;
        if(a[mid] == key)
        {
            found = 1;
            break;
        }
        else if(key < a[mid])
        {
            high = mid - 1;
        }
    }
```



```
else {  
    low = mid + 1;  
}  
}  
if(found == 1)  
{  
    printf("Element found at position %d",mid + 1);  
}  
else  
{  
    printf("Element not found");  
}  
getch();  
}
```

OUTPUT

```
Enter number of elements: 7
```

```
Enter 7 elements:
```

```
31
```

```
13
```

```
67
```

```
15
```

```
9
```

```
32
```

```
0
```

```
Sorted array:
```

```
0 9 13 15 31 32 67 _
```

PROGRAM – 33

AIM : Program to sort N elements in an array using bubble sort.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main() {
    int a[50], n, i, j, temp;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(a[j] > a[j + 1]) {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }
    printf("Sorted array:\n");
    for(i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    getch();
}
```

OUTPUT

```
Enter first string: Hello
Enter second string: World
Length of first string: 15597573
Copied string: Hello
Concatenated string: HelloWorld
Second string and copied string are not equal
Reversed second string: dlrow
```


PROGRAM – 34

AIM : Program to perform various string handling functions.

SOURCE CODE:

```
#include <stdio.h>
#include <string.h>
void main()
{
    char str1[50], str2[50], str3[50];
    int cmp;
    printf("Enter first string: ");
    scanf("%s", str1);
    printf("Enter second string: ");
    scanf("%s", str2);

    // 1. Length of string
    printf("Length of first string: %lu\n", strlen(str1));

    // 2. Copy string
    strcpy(str3, str1);
    printf("Copied string: %s\n", str3);

    // 3. Concatenate strings
    strcat(str1, str2);
    printf("Concatenated string: %s\n", str1);

    // 4. Compare strings
    cmp = strcmp(str2, str3);
    if (cmp == 0)
        printf("Second string and copied string are equal\n");
    else
        printf("Second string and copied string are not equal\n");
}
```



```
// 5. Reverse string (Turbo C)
strrev(str2);
printf("Reversed second string: %s\n", str2);
getch();
}
```

OUTPUT

```
Enter a string: madam  
The string is a Palindrome_
```

```
Enter a string: World  
The string is NOT a Palindrome
```

PROGRAM – 35

AIM : Program to read a string and check whether it is palindrome or not.

SOURCE CODE:

```
#include<stdio.h>
#include<string.h>
void main()
{
    char str[100];
    int i, len, flag = 0;
    clrscr();
    printf("Enter a string: ");
    scanf("%s", str);
    len = strlen(str);
    for (i = 0; i < len / 2; i++)
    {
        if (str[i] != str[len - i - 1])
        {
            flag = 1;
            break;
        }
    }
    if (flag == 0)
        printf("The string is a Palindrome");
    else
        printf("The string is NOT a Palindrome");
    getch();
}
```

OUTPUT

```
Enter a string: Hello  
Length of the string = 5_
```

```
Enter a string: Notebook  
Length of the string = 8
```

PROGRAM – 36

AIM : Program to find the length of a string without using the string handling function strlen().

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    char str[100];
    int length = 0;
    clrscr();
    printf("Enter a string: ");
    scanf("%s", str);
    while (str[length] != '\0')
    {
        length++;
    }
    printf("Length of the string = %d", length);
    getch();
}
```

OUTPUT

```
Enter a string: Hello World  
Original string: Hello World  
Copied string  : Hello World_
```

```
Enter a string: FRIDAY12345  
Original string: FRIDAY12345  
Copied string  : FRIDAY12345_
```


PROGRAM – 37

AIM : Program to copy one string to another without using string handling function.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main() {
    char str1[100], str2[100];
    int i = 0;
    printf("Enter a string: ");
    gets(str1);

    while(str1[i] != '\0') {
        str2[i] = str1[i];
        i++;
    }
    str2[i] = '\0';
    printf("Original string: %s\n", str1);
    printf("Copied string : %s", str2);
    getch();
}
```

OUTPUT

```
Enter first string: Hello  
Enter second string: World!!!!  
Concatenated string: Hello World!!!!
```

PROGRAM – 38

AIM : Program to concatenate one string to another without using the string handling function.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    char str1[100], str2[100];
    int i = 0, j = 0;
    printf("Enter first string: ");
    gets(str1); // For Turbo C++
    printf("Enter second string: ");
    gets(str2);

    while(str1[i] != '\0') {
        i++;
    }
    while(str2[j] != '\0') {
        str1[i] = str2[j];
        i++;
        j++;
    }
    str1[i] = '\0';
    printf("Concatenated string: %s", str1);
    getch();
}
```

OUTPUT

```
Enter first string: hello  
Enter second string: hello  
Strings are equal
```

```
Enter first string: hello!!  
Enter second string: hello  
Strings are not equal_
```

PROGRAM – 39

AIM : Program to compare two strings without using string handling function.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main() {
    char str1[100], str2[100];
    int i = 0, flag = 0;
    clrscr();
    printf("Enter first string: ");
    gets(str1);
    printf("Enter second string: ");
    gets(str2);
    while(str1[i] != '\0' && str2[i] != '\0') {
        if(str1[i] != str2[i]) {
            flag = 1;
            break;
        }
        i++;
    }
    if(str1[i] != str2[i])
        flag = 1;
    if(flag == 0)
        printf("Strings are equal");
    else
        printf("Strings are not equal");
    getch();
}
```

OUTPUT

```
Enter number of strings: 5
Enter 5 strings:
Pineapple
Orange
Apple
Kiwi
Grapes
Sorted strings:
Apple
Grapes
Kiwi
Orange
Pineapple
```

PROGRAM – 40

AIM : Program to sort N strings.

SOURCE CODE:

```
#include <stdio.h>
#include <string.h>
void main() {
    char str[20][50], temp[50];
    int n, i, j;
    clrscr();
    printf("Enter number of strings: ");
    scanf("%d", &n);
    printf("Enter %d strings:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%s", str[i]);
    }
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(strcmp(str[i], str[j]) > 0) {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }
    printf("Sorted strings:\n");
    for(i = 0; i < n; i++) {
        printf("%s\n", str[i]);
    }
    getch();
}
```

OUTPUT

```
Enter 1st numbers:    111
Enter 2nd numbers:    222

Before swapping: A = 111, B = 222
===== After swapByValue =====
  A = 222, B = 111
===== After swapByReference =====
  A = 222, B = 111
```


PROGRAM – 41

AIM : Program to swap two numbers using call by value and call by reference.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
int swapByValue(int a, int b)
{
    int temp = a;
    a = b;
    b = temp;
    printf("==== After swapByValue ==== \n A = %d, B = %d \n", a, b);
}
void swapByReference(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
    printf("==== After swapByReference ==== \n A = %d, B = %d \n", *a, *b);
}
void main()
{
    int a, b;
    clrscr();
    printf("Enter 1st numbers: \t");
    scanf("%d", &a);
    printf("Enter 2nd numbers: \t");
    scanf("%d",&b);
    printf("\nBefore swapping: A = %d, B = %d \n", a, b);
    swapByValue(a, b);
    swapByReference(&a, &b);
    getch();
}
```

OUTPUT

```
Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter elements of first matrix:
1
2
3
4
Enter elements of second matrix:

4
3
2
1
Resultant matrix:
8 5
20 13
```

PROGRAM – 42

AIM : Program to perform matrix multiplication using pointers.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main() {
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2;
    int i, j, k;
    printf("Enter rows and columns of first matrix: ");
    scanf("%d %d", &r1, &c1);
    printf("Enter rows and columns of second matrix: ");
    scanf("%d %d", &r2, &c2);

    // Matrix multiplication condition
    if(c1 != r2) {
        printf("Matrix multiplication not possible");
        return 0;
    }
    printf("Enter elements of first matrix:\n");
    for(i = 0; i < r1; i++) {
        for(j = 0; j < c1; j++) {
            scanf("%d", (*(a + i) + j)); // pointer access
        }
    }

    printf("Enter elements of second matrix:\n");
    for(i = 0; i < r2; i++) {
        for(j = 0; j < c2; j++) {
            scanf("%d", (*(b + i) + j)); // pointer access
        }
    }
}
```



```

// Matrix multiplication logic
for(i = 0; i < r1; i++) {
    for(j = 0; j < c2; j++) {
        *(*(c + i) + j) = 0;
        for(k = 0; k < c1; k++) {
            *(*(c + i) + j) +=
                (*(*(a + i) + k)) * (*(*(b + k) + j));
        }
    }
}

printf("Resultant matrix:\n");
for(i = 0; i < r1; i++) {
    for(j = 0; j < c2; j++) {
        printf("%d ", *(*(c + i) + j));
    }
    printf("\n");
}

getch();
}

```

OUTPUT

```
----- Vowels -----  
a,e,i,o,u  
A,E,I,O,U  
  
Enter a string:      Hello,Good Morning To All  
  
Number of vowels = 8_
```

PROGRAM – 43

AIM : Program to read a string and calculate the number of vowels in it.

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    char str[100];
    int i, count = 0;
    printf("----- Vowels ----- \n a,e,i,o,u \n A,E,I,O,U \n");
    printf("\n");
    printf("Enter a string: \t");
    gets(str);

    for(i = 0; str[i] != '\0'; i++) {
        if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
           str[i] == 'o' || str[i] == 'u' ||
           str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
           str[i] == 'O' || str[i] == 'U') {
            count++;
        }
    }
    printf("\n");
    printf("Number of vowels = %d", count);
    getch();
}
```

OUTPUT

```
Enter number of people: 3

Person 1
Enter name: Anu
Enter address: India
Enter phone number: 1234567891

Person 2
Enter name: Max
Enter address: America
Enter phone number: 2123456789

Person 3
Enter name: John
Enter address: Canada
Enter phone number: 4123567891

Enter name to search: Max

--- Person Found ---
Name      : Max
Address   : America
Phone     : 2123456789
```

```
Enter number of people: 3

Person 1
Enter name: John
Enter address: America
Enter phone number: 6789123456

Person 2
Enter name: Adam
Enter address: UK
Enter phone number: 4517289101

Person 3
Enter name: Anu
Enter address: Kerala
Enter phone number: 9124258912

Enter name to search: Manu

Person not found
```


PROGRAM – 44

AIM : Program to read the personal data (name, address, phone no etc...) of N people and search the details of a particular person using structures and functions

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>

struct person {
    char name[50];
    char address[100];
    char phone[15];
};

void readData(struct person p[], int n)
{
    int i;
    for(i = 0; i < n; i++)
    {
        printf("\nPerson %d\n", i + 1);
        printf("Enter name: ");
        gets(p[i].name);
        printf("Enter address: ");
        gets(p[i].address);
        printf("Enter phone number: ");
        gets(p[i].phone);
    }
}
```



```

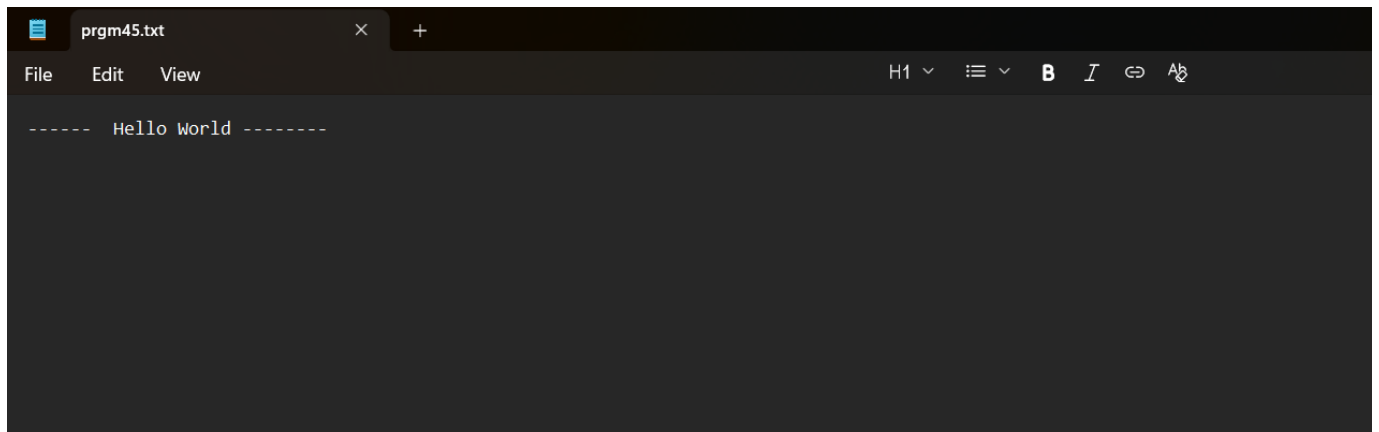
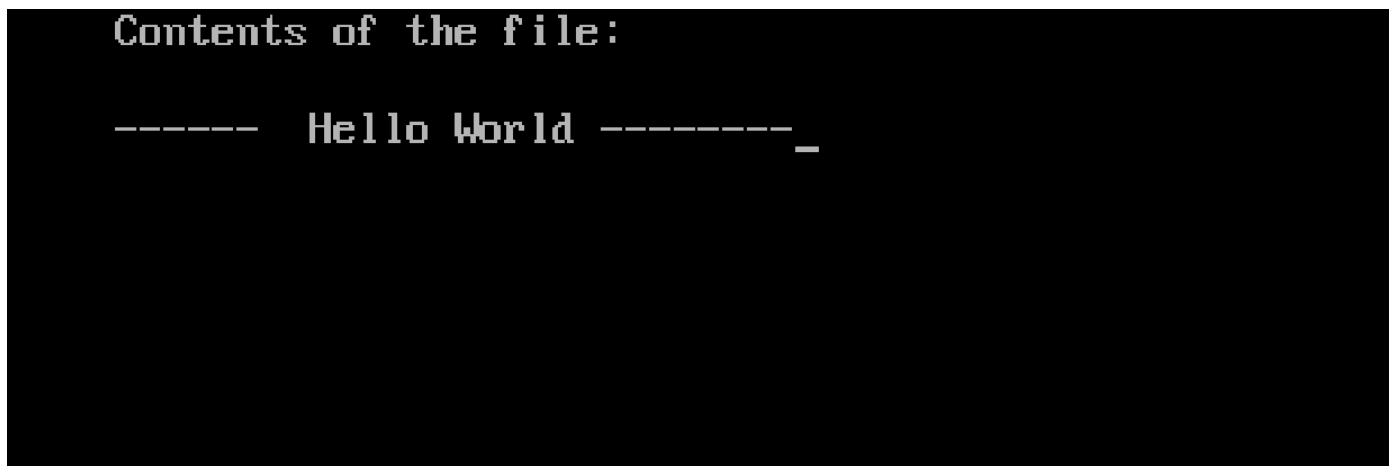
void searchData(struct person p[], int n)
{
    char searchName[50];
    int i, found = 0;
    printf("\nEnter name to search: ");
    gets(searchName);
    for(i = 0; i < n; i++)
    {
        if(strcmp(p[i].name, searchName) == 0)
        {
            printf("\n--- Person Found ---\n");
            printf("Name   : %s\n", p[i].name);
            printf("Address : %s\n", p[i].address);
            printf("Phone  : %s\n", p[i].phone);
            found = 1;
            break;
        }
    }
    if(found == 0)
        printf("\nPerson not found");
}

void main() {
    struct person p[20];
    int n;
    printf("Enter number of people: ");
    scanf("%d", &n);
    getchar();
    readData(p, n);
    searchData(p, n);
    getch();
}

```

OUTPUT

Create a file “prgm45.txt” and enter the content.

A screenshot of a text editor window with a dark theme. The title bar shows 'prgm45.txt' and a plus sign for a new file. The menu bar includes 'File', 'Edit', and 'View'. The toolbar on the right has icons for H1, list, bold, italic, link, and undo. The text area contains the line: '----- Hello world -----'.A screenshot of a terminal window with a black background and white text. The first line reads 'Contents of the file:'. The second line shows the output of a command, displaying '----- Hello World -----' followed by a cursor.

PROGRAM – 45

AIM : Program to read and display the contents of a file.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main() {
    FILE *fp;
    char ch;
    clrscr();

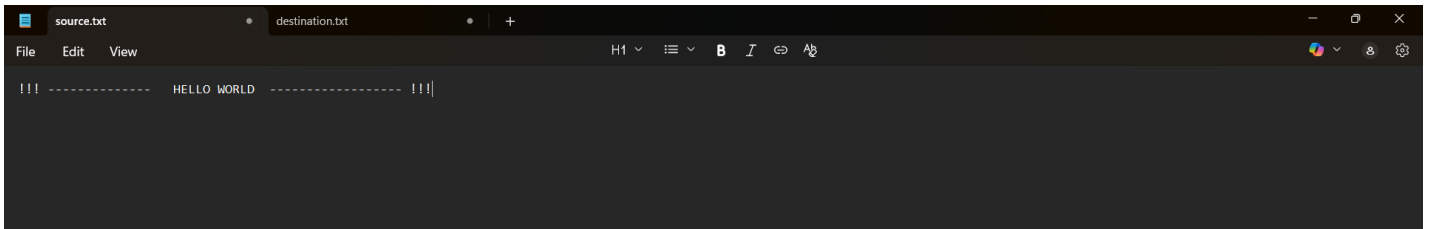
    fp = fopen("prgm45.txt", "r");

    if(fp == NULL) {
        printf("File not found or cannot be opened");
        getch();
        return;
    }
    printf("Contents of the file:\n\n");
    while((ch = fgetc(fp)) != EOF) {
        printf("%c", ch);
    }
    fclose(fp);
    getch();
}
```

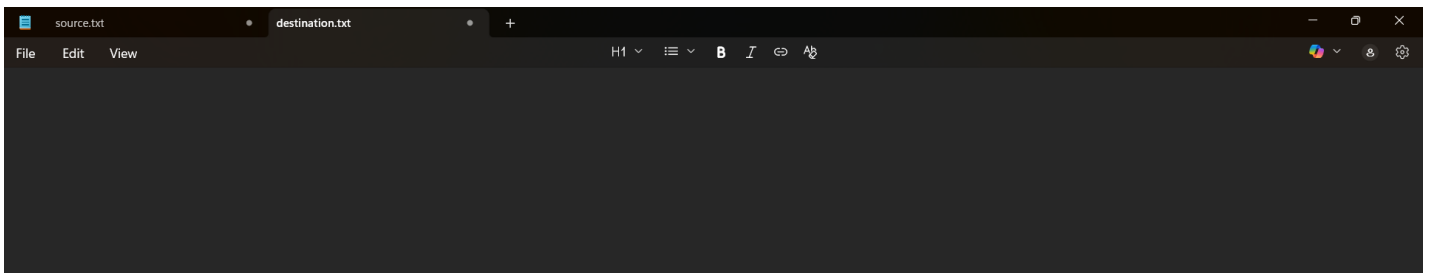
OUTPUT

Create two files “source.txt” & “destination.txt”.

Enter contents in “source.txt” & leave “destination.txt” as an empty file.



A screenshot of a code editor window with two tabs: 'source.txt' and 'destination.txt'. The 'source.txt' tab is active and shows the text '!!! ----- HELLO WORLD ----- !!!|'. The 'destination.txt' tab is empty. The editor has a dark theme and a menu bar with 'File', 'Edit', and 'View'.



A screenshot of a code editor window with two tabs: 'source.txt' and 'destination.txt'. The 'destination.txt' tab is active and shows the text '!!! ----- HELLO WORLD ----- !!!|'. The 'source.txt' tab is empty. The editor has a dark theme and a menu bar with 'File', 'Edit', and 'View'.

PROGRAM – 46

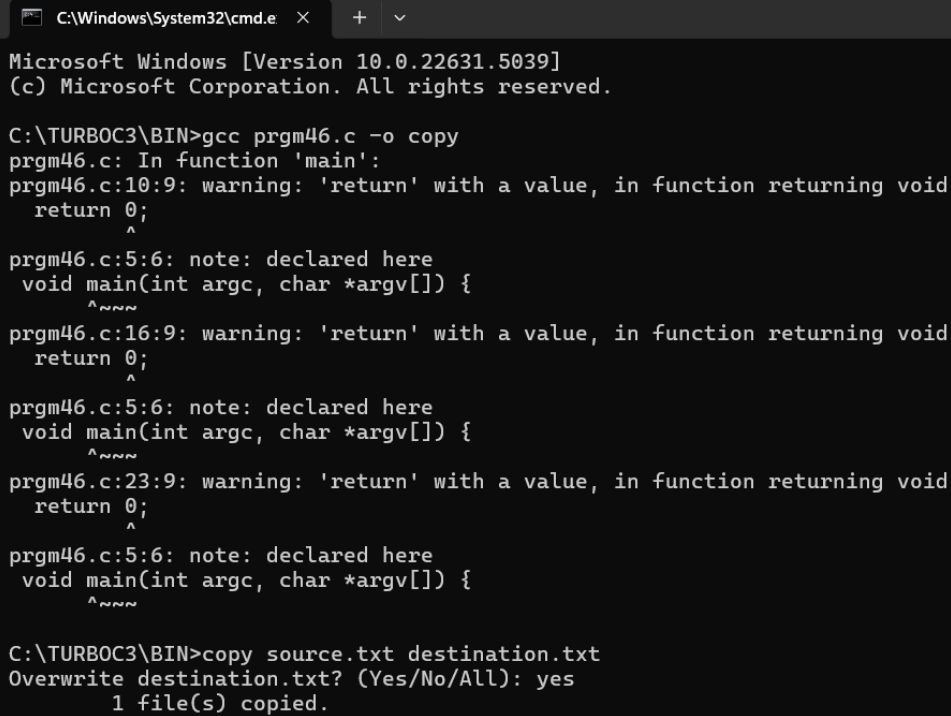
AIM : Program to copy one file to another using command line argument.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main(int argc, char *argv[]) {
    FILE *fs, *ft;
    char ch;
    if(argc != 3) {
        printf("Usage: copy source_file destination_file");
        return 0;
    }
    fs = fopen(argv[1], "r");
    if(fs == NULL) {
        printf("Source file cannot be opened");
        return 0;
    }
    ft = fopen(argv[2], "w");
    if(ft == NULL) {
        printf("Destination file cannot be created");
        fclose(fs);
        return 0;
    }
    while((ch = fgetc(fs)) != EOF) {
        fputc(ch, ft);
    }
```

OUTPUT

Open terminal from where the program is stored, type “gcc prgm46.c -o copy” in terminal.
Then, type “copy source.txt destination.txt”.

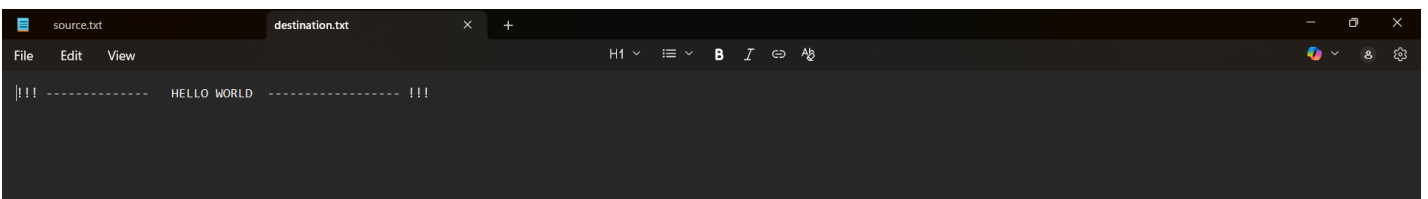


```
C:\Windows\System32\cmd.e  x  +  v

Microsoft Windows [Version 10.0.22631.5039]
(c) Microsoft Corporation. All rights reserved.

C:\TURBOC3\BIN>gcc prgm46.c -o copy
prgm46.c: In function 'main':
prgm46.c:10:9: warning: 'return' with a value, in function returning void
    return 0;
        ^
prgm46.c:5:6: note: declared here
    void main(int argc, char *argv[]) {
    ^~~~~
prgm46.c:16:9: warning: 'return' with a value, in function returning void
    return 0;
        ^
prgm46.c:5:6: note: declared here
    void main(int argc, char *argv[]) {
    ^~~~~
prgm46.c:23:9: warning: 'return' with a value, in function returning void
    return 0;
        ^
prgm46.c:5:6: note: declared here
    void main(int argc, char *argv[]) {
    ^~~~~

C:\TURBOC3\BIN>copy source.txt destination.txt
Overwrite destination.txt? (Yes/No/All): yes
    1 file(s) copied.
```



```
source.txt  destination.txt  x  +
File  Edit  View  H1  v  B  I  A

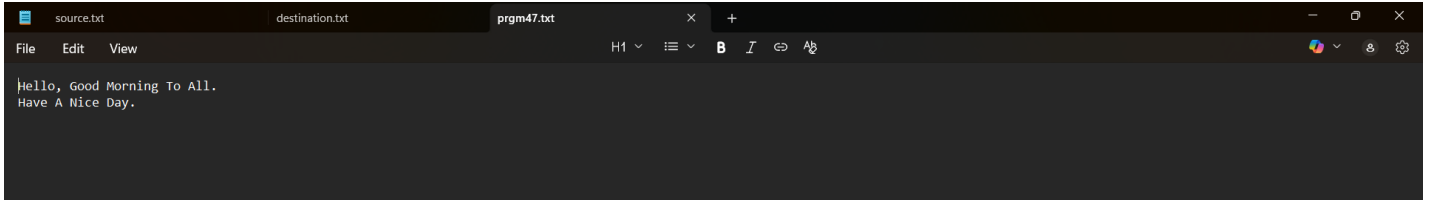
|!!  -----  HELLO WORLD  -----  !!|
```



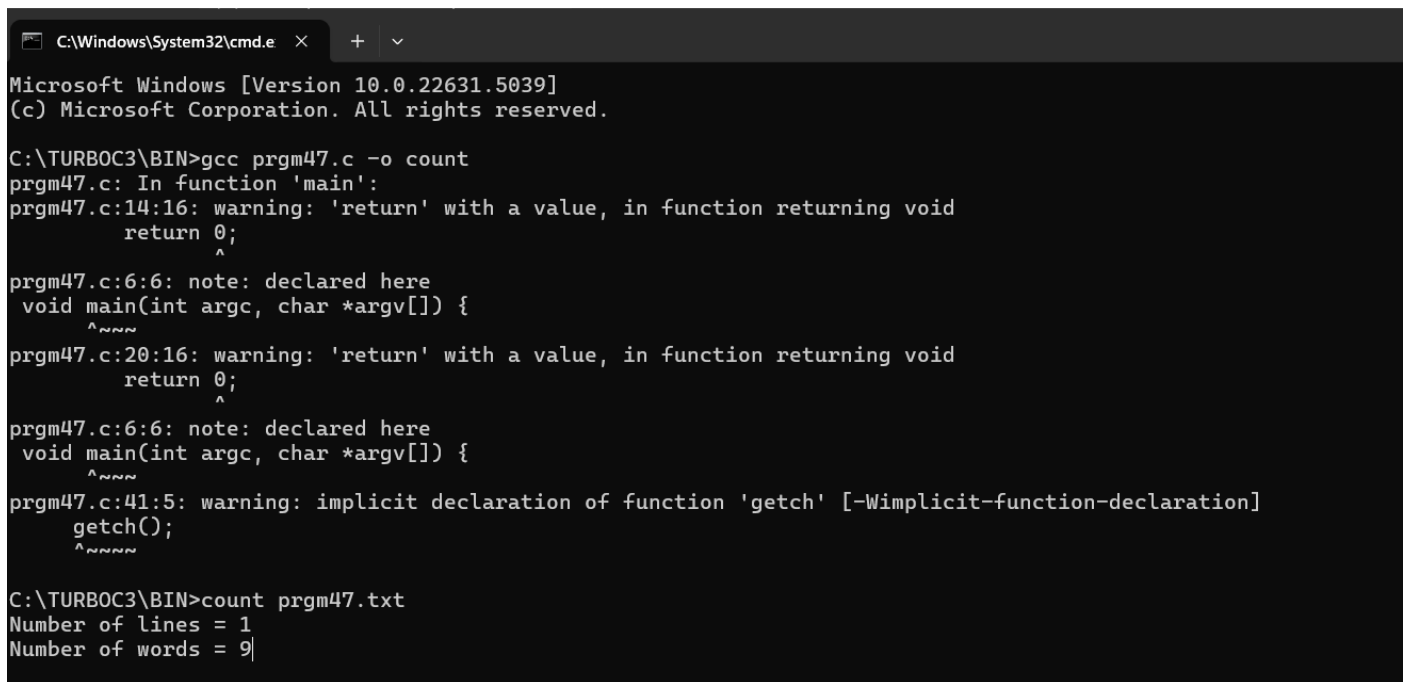
```
printf("File copied successfully");  
fclose(fs);  
fclose(ft);  
getch();  
}
```

OUTPUT

Create a file “prgm47.txt” and enter contents in it.

A screenshot of a text editor window with three tabs: 'source.txt', 'destination.txt', and 'prgm47.txt'. The 'prgm47.txt' tab is active, showing the text: 'Hello, Good Morning To All.' followed by 'Have A Nice Day.' on the next line. The editor has a dark theme and standard Windows-style window controls.

Open terminal from where the program is stored, type “gcc prgm47.c -o count” in terminal. Then, type “count prgm47.txt”.

A screenshot of a Windows command prompt window. The title bar shows 'C:\Windows\System32\cmd.e'. The prompt is 'Microsoft Windows [Version 10.0.22631.5039] (c) Microsoft Corporation. All rights reserved.' The user enters 'C:\TURBOC3\BIN>gcc prgm47.c -o count'. The compiler outputs several warnings and notes: 'prgm47.c: In function 'main':', 'prgm47.c:14:16: warning: 'return' with a value, in function returning void', 'prgm47.c:6:6: note: declared here', 'prgm47.c:20:16: warning: 'return' with a value, in function returning void', 'prgm47.c:6:6: note: declared here', and 'prgm47.c:41:5: warning: implicit declaration of function 'getch' [-Wimplicit-function-declaration]'. The user then enters 'C:\TURBOC3\BIN>count prgm47.txt', and the output shows 'Number of lines = 1' and 'Number of words = 9'.

PROGRAM – 47

AIM: Program to count the number of words and number of lines in a file using command line argument.

SOURCE CODE:

```
#include <stdio.h>
#include <stdlib.h>
void main(int argc, char *argv[]) {
    FILE *fp;
    char ch;
    int words = 0, lines = 0;
    int inWord = 0;
    if (argc != 2) {
        printf("Invalid arguments");
        return 0;
    }
    fp = fopen(argv[1], "r");
    if (fp == NULL) {
        printf("File cannot be opened");
        return 0;
    }

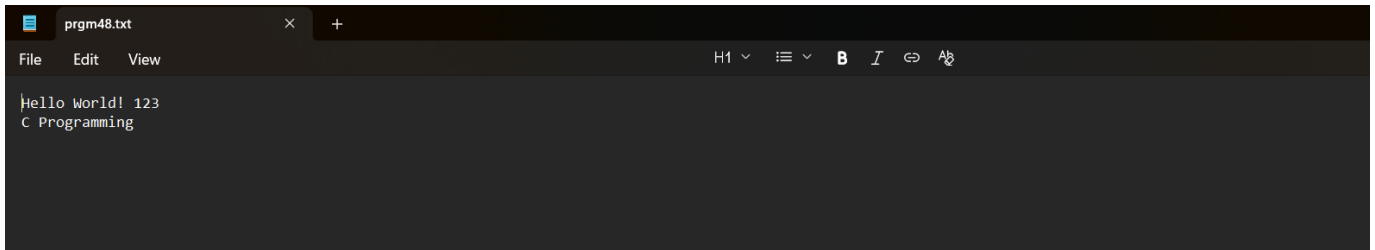
    while ((ch = fgetc(fp)) != EOF) {
        if (ch == '\n')
            lines++;
        if (ch == ' ' || ch == '\n' || ch == '\t')
            inWord = 0;
        else if (inWord == 0) {
            inWord = 1;
            words++;
        }
    }
}
```



```
fclose(fp);  
printf("Number of lines = %d\n", lines);  
printf("Number of words = %d", words);  
getch();  
}
```

OUTPUT

Create a file “prgm48.txt” and enter contents in it.

A screenshot of a text editor window with a dark theme. The title bar shows 'prgm48.txt' and a '+' icon. The menu bar includes 'File', 'Edit', and 'View'. The toolbar shows 'H1', a list icon, 'B' (bold), 'I' (italic), '↔' (undo), and a redo icon. The text area contains two lines: 'Hello World! 123' and 'C Programming'.

Contents of the file:

```
Hello World! 123
C Programming
```

```
Lowercase characters: 18
Uppercase characters: 4
Special characters: 7_
```

PROGRAM – 48

AIM : Program to count the number of lower case, upper case and special characters in a file

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>

void main() {
    FILE *fp;
    char ch;
    int lower = 0, upper = 0, special = 0;
    clrscr();
    fp = fopen("prgm48.txt", "r");
    if(fp == NULL) {
        printf("File cannot be opened");
        getch();
    }
    printf("Contents of the file:\n\n");
    while((ch = fgetc(fp)) != EOF) {
        printf("%c", ch);
        if(ch >= 'a' && ch <= 'z')
            lower++;
        else if(ch >= 'A' && ch <= 'Z')
            upper++;
        else if(ch != '\n' && ch != '\r') // Ignore newline
            special++;
    }
    fclose(fp);
    printf("\n\nLowercase characters: %d\n", lower);
    printf("Uppercase characters: %d\n", upper);
    printf("Special characters: %d", special);
    getch();
}
```

OUTPUT

```
Enter number of employees: 3
```

```
Employee 1 details:
```

```
ID: 121
```

```
Name: Anu
```

```
Designation: Staff
```

```
Salary: 8000
```

```
Employee 2 details:
```

```
ID: 101
```

```
Name: Manu
```

```
Designation: Manager
```

```
Salary: 8500
```

```
Employee 3 details:
```

```
ID: 291
```

```
Name: Jobin
```

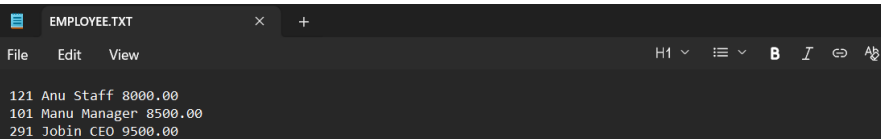
```
Designation: CEO
```

```
Salary: 9500
```

```
Employee Details:
```

ID	Name	Designation	Salary
121	Anu	Staff	8000.00
101	Manu	Manager	8500.00
291	Jobin	CEO	9500.00

```
_
```



```
EMPLOYEE.TXT
File Edit View H1 B I U A
121 Anu Staff 8000.00
101 Manu Manager 8500.00
291 Jobin CEO 9500.00
```


PROGRAM – 49

AIM : Create an employee file, read and display the details on console.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
struct employee {
    int id;
    char name[50];
    char designation[50];
    float salary;
};
void main() {
    FILE *fp;
    struct employee emp;
    int n, i;
    clrscr();
    printf("Enter number of employees: ");
    scanf("%d", &n);
    getchar();
    fp = fopen("employee.txt", "w");
    if(fp == NULL) {
        printf("Cannot create file");
        getch();
        return 0;
    }
```



```

// Input employee details and write to file
for(i = 0; i < n; i++) {
    printf("\nEmployee %d details:\n", i + 1);
    printf("ID: ");
    scanf("%d", &emp.id);
    getchar(); // clear buffer

    printf("Name: ");
    gets(emp.name);

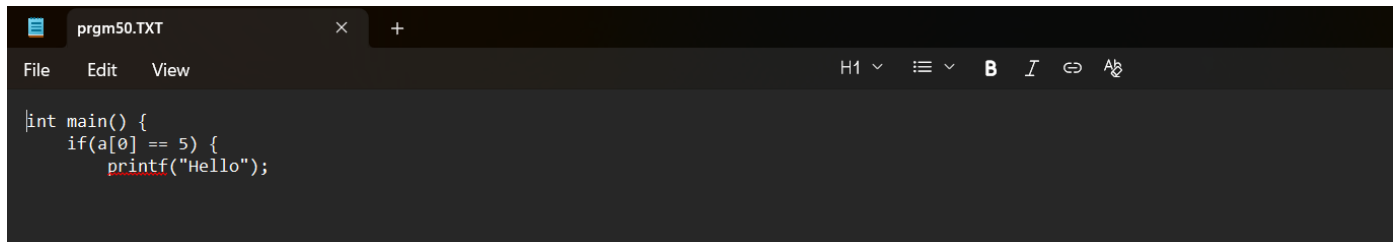
    printf("Designation: ");
    gets(emp.designation);

    printf("Salary: ");
    scanf("%f", &emp.salary);
    getchar();
    fprintf(fp, "%d %s %s %.2f\n", emp.id, emp.name, emp.designation, emp.salary);
}
fclose(fp);
fp = fopen("employee.txt", "r");
if(fp == NULL) {
    printf("Cannot open file");
    getch();
}
printf("\nEmployee Details:\n");
printf("ID\tName\tDesignation\tSalary\n");

while(fscanf(fp, "%d %s %s %f", &emp.id, emp.name, emp.designation, &emp.salary) != EOF) {
    printf("%d\t%s\t%s\t%.2f\n", emp.id, emp.name, emp.designation, emp.salary);
}
fclose(fp);
getch();
}

```

OUTPUT



A screenshot of a code editor window titled 'prgm50.TXT'. The editor has a menu bar with 'File', 'Edit', and 'View'. The right side of the menu bar contains formatting options: 'H1', a list icon, 'B' (bold), 'I' (italic), '↺' (undo), and '↻' (redo). The code in the editor is:

```
int main() {  
    if(a[0] == 5) {  
        printf("Hello");  
    }
```

Contents of the file:

```
int main() {  
    if(a[0] == 5) {  
        printf("Hello");  
    }
```

Bracket Mismatch Check:

Curly brackets {} are mismatched

PROGRAM – 50

AIM : Program to find the mismatch of opening and closing brackets in a file.

SOURCE CODE:

```
#include <stdio.h>
#include <conio.h>
void main() {
    FILE *fp;
    char ch;
    int round = 0, curly = 0, square = 0;
    clrscr();
    fp = fopen("prgm50.txt", "r");
    if(fp == NULL) {
        printf("Cannot open file");
        getch();
        return;
    }
    printf("Contents of the file:\n\n");
    // Read file and display content
    while((ch = fgetc(fp)) != EOF) {
        printf("%c", ch);
        // Count brackets
        if(ch == '(') round++;
        else if(ch == ')') round--;
        else if(ch == '{') curly++;
        else if(ch == '}') curly--;
        else if(ch == '[') square++;
        else if(ch == ']') square--;
    }
    fclose(fp);
}
```



```
printf("\n\nBracket Mismatch Check:\n");
if(round == 0 && curly == 0 && square == 0)
    printf("All brackets are matched\n");
else {
    if(round != 0) printf("Round brackets () are mismatched\n");
    if(curly != 0) printf("Curly brackets {} are mismatched\n");
    if(square != 0) printf("Square brackets [] are mismatched\n");
}
getch();
}
```


C PROJECT

BANKING MANAGEMENT SYSTEM

OUTPUT

[illegible]

```

-----Main Menu-----
|
| 1. Manage Accounts
| 2. Manage Transactions
| 3. Manage Loans
|
| 0. Exit
|
|
|
|
| Enter Choice...
|
|
|
|
|
|
|
|
-----

```

SOURCE CODE:

```
// Declaring header
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include<conio.h>

#define yellow "\033[33m"
#define cyan "\033[36m"
#define red "\033[31m"
#define white "\033[37m"

#define width 80
#define height 22

int op, choice, nav, count, y, page;
// Declaring struct
// Account Structure
struct account{
    int a_id;
    char a_name[20];
    char a_addr[40];
    int a_bal;
}a1;
// Transaction Structure
struct transaction{
    int t_id;
    char a_no[20];
    char t_date[12];
    int t_type;
    int t_amt;
}t1;
```

OUTPUT

```

-----ACCOUNT MENU-----
|
|
|      1. Add Account
|      2. Edit Account
|      3. Delete Account
|      4. View Account
|
|      9. Back
|      0. Exit
|      Enter Choice...
|
|      -
|
|
|
|
|
|
|
|
-----

```

```

-----Create Account-----
|
|   Enter Account Number: 121
|
|   Enter Account Holder Name: Pinto
|
|   Enter Address: MACFAST
|
|   Enter Balance: 2500
|
|
|
|
|   Account Created Successfully.
|
|
|
|
-----

```

```

// Loan Structure
struct loan{
    int l_id;
    int a_no;
    char l_date[12];
    char l_type[20];
    int l_amt;
    int l_dur;
}11;

// Declaration

// Menu
int operationMenu();
int manageAccount();
int manageTransaction();
int manageLoan();

// Creations
int createAccount();
int createTransaction();
int createLoan();

// Edit
int editAccount();
int editTransaction();
int editLoan();

// Delete
int deleteAccount();
int deleteTransaction();
int deleteLoan();

// View
int viewAccount();
int viewTransaction();
int viewLoan();

//utilize
void gotoxy( int, int);

```

OUTPUT

-----Edit Account

Enter Account ID: 121

New Account Name: **Printoo**

New Address: **MAC**

New Balance: 3500

Accounts Updated!

Press Enter to continue...

-View Account:

Accounts List...

Acc: 1232 Name: Rahul MACFAST Add: QWRTETYE HHJ Bal: 2000

Acc: 121 Name: Printoo Add: MAC Bal: 3500

```

void delay();
void tuiFrame();
void drawLine(char, int);
void clearInside();
void clearScreen();
void renderScreen(const char *title);
void screenMainMenu();
int nextButton();
int exiting();
int bottomBar();
void screenAccounts();
void screenTransaction();
void screenLoan();
//move cursor
void gotoXY(int x, int y) {
    printf("\033[%d;%dH", y, x); // \033 ansi value of escape, H move cursor.
}
//delayer
void delay(){
    gotoXY(52, height);
    printf(cyan"Press any key to continue...");
    gotoXY(79, height);
    getch();
    return;
}
// Styling
void tuiFrame() {
    int i,j;
    // TOP BORDER
    for(i = 0; i < width; i++) printf(cyan"-");
    printf("\n");
    // SIDES
    for(i = 0; i < height - 2; i++) {
        printf(cyan"|");
    }
}

```

OUTPUT

```

-----Delete Account-----
|
|   Enter Account ID to Delete: 121
|
|
|
|   Deleted!
|
|
|
|
|
|
|
|
|
|
-----

```

```
-----View Account-----  
:  
  
Accounts List...  
:  
Acc: 1232 Name: Rahul MACFAST Add: QWRTETYE HHJ Bal: 2000 _  
:  
:  
:  
:  
:  
:  
:  
:  
:  
:  
:  
:
```



```

        for(j = 0; j < width - 2; j++) {
            printf(" ");
        }
        printf(cyan"|\n");
    }
    // BOTTOM BORDER
    for(i = 0; i < width; i++) printf("-");
    printf("\n");
}

void drawLine(char ch, int limit){
    int i;
    for(i = 0; i < limit; i++){
        printf(cyan"%c", ch);
    }
}

// Clear Inside
void clearInside() {
    int x,y;
    for (y = 3; y <= 22; y++) { // Leave title line intact
        gotoXY(2, y);
        for (x = 0; x < 76; x++)
            printf(" ");
    }
}

void clearScreen() {
    printf("\033[2J\033[1;1H");
}

void renderScreen(const char *title){
    int len = strlen(title);
    int y = 0;
    y = (80 - len)/2;
    tuiFrame();
    gotoXY(1,1);
    drawLine('-', 80);    // Draw the fixed 80x22 frame

```

OUTPUT

TRANSACTION MENU

```

1. Add Transaction
2. Edit Transaction
3. Delete Transaction
4. View Transaction

9. Back
0. Exit
Enter Choice...

```

-Create Transaction

```
Enter Transaction ID: 101
Enter Account Number: 1234
Enter Transaction Date (DD-MM-YYYY): 01-12-1998
Enter Transaction Type (enter no. 1 = Debit, 2 = Credit): 2
Enter Transaction Amount: 500
```

Transaction Data Added Successfully.

```

gotoXY(1,2);
drawLine('-', 80);
gotoXY(y,2);    // Put title inside the box
printf("%s", title );
printf(cyan);
clearInside();  // Clear the content area for next screen
}
void screenMainMenu() {
    renderScreen(yellow"Main Menu"); // clears and draws frame
    gotoXY(10, 4); printf(white"1. Manage Accounts");
    gotoXY(10, 5); printf("2. Manage Transactions");
    gotoXY(10, 6); printf("3. Manage Loans");
    gotoXY(10, 8); printf("0. Exit");
}
int nextButton(){
    return 0;
}
int exiting(){
    gotoXY(39,20);
    printf(red"Exiting....");
    exit(0);
    return 0;
}
int bottomBar() {
    int op;
    gotoXY(10,20);
    printf("9. Exit");
    gotoXY(71,20);
    printf("0. Next");
    gotoXY(39,20);
    scanf("%d", &op);
    switch (op) {
        case 0: return 0; // next page
        case 9: return 9; // exit
    }
}

```

OUTPUT

```
-----Edit Transaction-----
Enter Transaction ID: 101
New Account Number: 1234
New Transaction Date: 03-09-1997
New Transaction Type: 2
New Transaction Amount: 300
Transaction Updated!

Press Enter to continue...
```

```

-----View Transaction-----
:
:
:   Transactions List...
:
:   Tran ID: 12345 Acc No: 2166 Tran Date: 01-07-2001 Tran Type: 2 Tran
Amt: 1500Tran ID: 1212 Acc No: 1500 Tran Date: 11-11-2011 Tran Type: 2 Tran A
mt: 1000 Tran ID: 101 Acc No: 1234 Tran Date: 03-09-1997 Tran Type: 2 Tran Am
t: 300 _
:
:
:
:
:
:
:
:
:
:

```

```

        default:
            gotoXY(30,20);
            printf("Enter valid choice...");
            getch();
            return bottomBar();
    }
}

void screenAccounts() {
    clearInside();
    gotoXY(30, 4); renderScreen(yellow"ACCOUNT MENU");
    gotoXY(10, 6); printf(white"1. Add Account");
    gotoXY(10, 7); printf("2. Edit Account");
    gotoXY(10, 8); printf("3. Delete Account");
    gotoXY(10, 9); printf("4. View Account");
    gotoXY(10, 11); printf("9. Back");
    gotoXY(10, 12); printf("0. Exit");
}

void screenTransaction() {
    clearInside();
    gotoXY(30, 4); renderScreen(yellow"TRANSACTION MENU");
    gotoXY(10, 6); printf(white"1. Add Transaction");
    gotoXY(10, 7); printf("2. Edit Transaction");
    gotoXY(10, 8); printf("3. Delete Transaction");
    gotoXY(10, 9); printf("4. View Transaction");
    gotoXY(10, 11); printf("9. Back");
    gotoXY(10, 12); printf("0. Exit");
}

void screenLoan() {
    clearInside();
    gotoXY(30, 4); renderScreen(yellow"LOAN MENU");
    gotoXY(10, 6); printf(white"1. Add Loan");
    gotoXY(10, 7); printf("2. Edit Loan");
    gotoXY(10, 8); printf("3. Delete Loan");
    gotoXY(10, 9); printf("4. View Loan");

```

OUTPUT

```

-----Delete Transaction-----
|
|   Enter Transaction ID to Delete: 101
|
|
|
|   Deleted!
|
|
|
|
|
|
|
|
|
|
-----

```

```

    gotoXY(10, 11); printf("9. Back");
    gotoXY(10, 12); printf("0. Exit");
}
// Main function
int main(){
    // Declaring Menu Option
    // This Will be the one function triggered
    // All the actions are followed
    renderScreen(yellow"Bank Management System");
    gotoXY(32, 11);
    printf("Welcome to Bank MIS");
    gotoXY(32, 12);
    printf("-----");
    gotoXY(59, 22);
    printf("Enter to continue...");
    getch();
    operationMenu();
    return 0;
}
// All Banking Operations
int operationMenu(){
    while(1){
        // A pause which hlp with a gap
        screenMainMenu();
        gotoXY(10,13);
        printf("Enter Choice...\n");
        gotoXY(14,14);
        scanf("%d", &choice);
        switch (choice){
            case 1:
                manageAccount();
                break;
            case 2:
                manageTransaction();

```

OUTPUT

```

1. Add Loan
2. Edit Loan
3. Delete Loan
4. View Loan

9. Back
0. Exit
Enter Choice...
```

```
Enter Loan ID: 111
Enter Account Number: 1234
Enter Loan Date (DD-MM-YYYY): 09-11-2012
Enter Loan Type (secure, unsecure, business): business
Enter Loan Amount: 20000
Enter Loan Duration: 5 years
Loan Data Added Successfully.
```



```

        break;
    case 3:
        manageLoan();
        break;
    case 0:
        exiting();
        break;
    default:
        printf("Invalid Choice.\n");
    }
}
}
// Insert Operation
int manageAccount()
{
    screenAccounts();
    gotoXY(10,13);
    printf("Enter Choice...\n");
    gotoXY(14,14);
    scanf("%d", &op);
    switch (op){
        case 1:
            createAccount();
            break;
        case 2:
            editAccount();
            break;
        case 3:
            deleteAccount();
            break;
        case 4:
            viewAccount();
            break;
        case 9:

```

OUTPUT

```

-----Edit Loan-----
Enter Loan ID: 111
New Account Number: 1234
New Loan Date: 11-10-2001
New Loan Type: 3
New Loan Amount: 7000
New Duration: 10
Updated!

Press Enter to continue...

```

```

-----View Loan-----
|
|      Loan List...
|
|      Loan ID: 111Acc No: 1234L Date:  11-10-2001L Type:  3L Amt: 7000Dur: 10
|
|
|
|
|
|
|
|
|
|
|
|

```

```

        operationMenu();
        break;
    case 0:
        exiting();
        break;
    default:
        printf("Invalid Operation.");
}
return 0;
}

// Edit Opeeration
int manageTransaction(){
    screenTransaction();
    gotoXY(10,13);
    printf("Enter Choice...\n");
    gotoXY(14,14);
    scanf("%d", &op);
    switch (op){
        case 1:
            createTransaction();
            break;
        case 2:
            editTransaction();
            break;
        case 3:
            deleteTransaction();
            break;
        case 4:
            viewTransaction();
            break;
        case 9:
            operationMenu();
            break;
        case 0:

```

OUTPUT

[illegible][illegible]

```

        exiting();
        break;
    default:
        printf("Invalid Operation.");
    }
    return 0;
}
// Delete Operation
int manageLoan(){
    screenLoan();
    gotoXY(10,13);
    printf("Enter Choice...\n");
    gotoXY(14,14);
    scanf("%d", &op);
    switch (op){
        case 1:
            createLoan();
            break;
        case 2:
            editLoan();
            break;
        case 3:
            deleteLoan();
            break;
        case 4:
            viewLoan();
            break;
        case 9:
            operationMenu();
            break;
        case 0:
            exiting();
            break;
    }
}

```



```

    default:

        printf("Invalid Operation.");

    }

    return 0;
}

// Validations for Unique ID
int validAcc(int id){
    FILE *fAcc = fopen("accounts.txt", "r");
    while (fscanf(fAcc, "%d|%^]|%[^]]%d\n",
        &a1.a_id, a1.a_name, a1.a_addr, &a1.a_bal) != EOF) {
        if(a1.a_id == id){
            fclose(fAcc);
            return 0;
        }
    }
    fclose(fAcc);
    return 1;
}

int validTran(int id){
    FILE *fTran = fopen("transaction.txt", "r");
    while (fscanf(fTran, "%d|%^]|%[^]]%d|%d\n",
        &t1.t_id, t1.a_no, t1.t_date,&t1.t_type, &t1.t_amt) != EOF) {
        if(t1.t_id == id){
            fclose(fTran);
            return 0;
        }
    }
    fclose(fTran);
    return 1;
}

int validLoan(int id){
    FILE *fLoan = fopen("loan.txt", "r");
    while (fscanf(fLoan, "%d|%d|%^]|%[^]]%d|%d\n", &l1.l_id, &l1.a_no, l1.l_date,l1.l_type, &l1.l_amt,

```



```

&l1.l_dur) != EOF) {
    if (l1.l_id == id){
        fclose(fLoan);
        return 0;
    }
}
fclose(fLoan);
return 1;
}

```

// FILE INSERTION OPERATIONS

// Inserting data into Accounts file

```

int createAccount() {
    int id;
    FILE *fAcc = fopen("accounts.txt", "a");
    if (fAcc == NULL) {
        gotoXY(10,4);
        printf(red"Error opening Accounts file.\n");
        return 1;
    }
    do{
        renderScreen("Create Account");
        gotoXY(10,4);
        printf(white"Enter Account Number: ");
        scanf("%d", &id);
        if(!validAcc(id)){
            gotoXY(10,6);
            printf(yellow"Account No. already exists! Enter another.");
            printf(white);
            delay();
        }
    }while (!validAcc(id));
    a1.a_id = id;
    gotoXY(10,6);
    printf("Enter Account Holder Name: ");

```



```

scanf("%[^\\n]", a1.a_name);
gotoXY(10,8);
printf("Enter Address: ");
scanf("%[^\\n]", a1.a_addr);
gotoXY(10,10);
printf("Enter Balance: ");
scanf("%d", &a1.a_bal);
fprintf(fAcc, "%d|%s|%s|%d\\n", a1.a_id, a1.a_name, a1.a_addr, a1.a_bal);
fclose(fAcc);
gotoXY(10,16);
printf(yellow"Account Created Successfully.\\n\\n");
delay();
return 0;
}

// Inserting data into Transactions file
int createTransaction() {
    int id;
    FILE *fTran = fopen("transaction.txt", "a");
    if (fTran == NULL) {
        gotoXY(10,4);
        printf(red"Error opening Transactions file.\\n");
        return 1;
    }
    do{
        renderScreen("Create Transaction");
        gotoXY(10,4);
        printf(white"Enter Transaction ID: ");
        scanf("%d", &id);
        if(!validTran(id)){
            gotoXY(10,6);
            printf(yellow"Transaction ID already exists! Enter another.");
            printf(white);
            delay();
        }
    }

```



```

    } while (!validTran(id));
    t1.t_id = id;
    gotoXY(10,6);
    printf("Enter Account Number: ");
    scanf(" %s", t1.a_no);
    gotoXY(10,8);
    printf("Enter Transaction Date (DD-MM-YYYY): ");
    scanf(" %s", t1.t_date);
    gotoXY(10,10);
    printf("Enter Transaction Type (enter no. 1 = Debit, 2 = Credit): ");
    scanf("%d", &t1.t_type);
    gotoXY(10,12);
    printf("Enter Transaction Amount: ");
    scanf("%d", &t1.t_amt);
    fprintf(fTran, "%d|s|s|d|d\n", t1.t_id, t1.a_no, t1.t_date, t1.t_type, t1.t_amt);
    fclose(fTran);
    gotoXY(10,16);
    printf(yellow"Transaction Data Added Successfully.\n\n");
    delay();
    return 0;
}

// Inserting data into Loan file
int createLoan() {
    int id;
    FILE *fLoan = fopen("loan.txt", "a");
    if (fLoan == NULL) {
        gotoXY(10,4);
        printf(red"Error opening Loans file.\n");
        return 1;
    }
    do{
        renderScreen("Create Loan");
        gotoXY(10,4);
        printf(white"Enter Loan ID: ");

```



```

scanf("%d", &id);
if(!validLoan(id)){
    gotoXY(10,6);
    printf(yellow"ID already exists! Enter another.");
    printf(white);
}
}while (!validLoan(id));
l1.l_id = id;
gotoXY(10,6);
printf("Enter Account Number: ");
scanf("%d", &l1.a_no);
gotoXY(10,8);
printf("Enter Loan Date (DD-MM-YYYY): ");
scanf(" %[^\\n]", l1.l_date);
gotoXY(10,10);
printf("Enter Loan Type (secure, unsecure, business): ");
scanf(" %[^\\n]", l1.l_type);
gotoXY(10,12);
printf("Enter Loan Amount: ");
scanf("%d", &l1.l_amt);
gotoXY(10,14);
printf("Enter Loan Duration: ");
scanf("%d", &l1.l_dur);
fprintf(fLoan, "%d|%d|%s|%s|%d|%d\\n",l1.l_id, l1.a_no, l1.l_date, l1.l_type, l1.l_amt, l1.l_dur);
fclose(fLoan);
gotoXY(10,16);
printf(yellow"Loan Data Added Successfully.\\n\\n");
getch();
return 0;
}

// FILE EDIT OPERATIONS
// Edit data in Account file
int editAccount() {
    FILE *fAcc = fopen("accounts.txt", "r");

```



```

FILE *temp = fopen("temp_acc.txt", "w");
int id, found = 0;
renderScreen("Edit Account");
gotoXY(10,4);
printf(white"Enter Account ID: ");
scanf("%d", &id);
while (fscanf(fAcc, "%d|%^]|%^]|%d\n",&a1.a_id, a1.a_name, a1.a_addr, &a1.a_bal) != EOF) {
    if (a1.a_id == id) {
        found = 1;
        gotoXY(10,6);
        printf("New Account Name: ");
        scanf(" %[^\\n]",a1.a_name);
        gotoXY(10,8);
        printf("New Address: ");
        scanf(" %[^\\n]",a1.a_addr);
        gotoXY(10,10);
        printf("New Balance: ");
        scanf("%d", &a1.a_bal);
    }
    fprintf(temp, "%d|s|s|d\\n",a1.a_id, a1.a_name, a1.a_addr, a1.a_bal);
}
fclose(fAcc);
fclose(temp);
remove("accounts.txt");
rename("temp_acc.txt", "accounts.txt");
if (found){
    gotoXY(10,12);
    printf(yellow"Accounts Updated!\\n");
}
else{
    gotoXY(10,12);
    printf(red"Id Not Found!\\n");
}
getch();

```



```

    return 0;
}
// Edit data in Transaction file
int editTransaction() {
    FILE *fTran = fopen("transaction.txt", "r");
    FILE *temp = fopen("temp_tran.txt", "w");
    int id, found = 0;
    renderScreen("Edit Transaction");
    gotoXY(10,4);
    printf(white"Enter Transaction ID: ");
    scanf("%d", &id);
    while (fscanf(fTran, "%d|^[^|]|^[^|]|%d|%d\n",&t1.t_id, t1.a_no, t1.t_date,&t1.t_type, &t1.t_amt) !=
EOF) {
        if (t1.t_id == id) {
            found = 1;
            gotoXY(10,6);
            printf("New Account Number: ");
            scanf(" %[^n]", t1.a_no);
            gotoXY(10,8);
            printf("New Transaction Date: ");
            scanf(" %[^n]", t1.t_date);
            gotoXY(10,10);
            printf("New Transaction Type: ");
            scanf("%d", &t1.t_type);
            gotoXY(10,12);
            printf("New Transaction Amount: ");
            scanf("%d", &t1.t_amt);
        }
        fprintf(temp, "%d|%s|%s|%d|%d\n",t1.t_id, t1.a_no, t1.t_date,t1.t_type, t1.t_amt);
    }
    fclose(fTran);
    fclose(temp);
    remove("transaction.txt");
    rename("temp_tran.txt", "transaction.txt");
}

```



```

if (found){
    gotoXY(10,14);
    printf(yellow"Transaction Updated!");
}
else{
    gotoXY(10,14);
    printf(red"Not Found!");
}
delay();
return 0;
}

// Edit data in Loan file
int editLoan() {
    FILE *fLoan = fopen("loan.txt", "r");
    FILE *temp = fopen("temp_loan.txt", "w");
    int id, found = 0;
    renderScreen("Edit Loan");
    gotoXY(10,4);
    printf(white"Enter Loan ID: ");
    scanf("%d", &id);
    while (fscanf(fLoan, "%d|%d|^[^|]|^[^|]|%d|%d\n",&l1.l_id, &l1.a_no, l1.l_date,l1.l_type, &l1.l_amt,
&l1.l_dur) != EOF) {
        if (l1.l_id == id) {
            found = 1;
            gotoXY(10,6);
            printf("New Account Number: ");
            scanf("%d", &l1.a_no);
            gotoXY(10,8);
            printf("New Loan Date: ");
            scanf(" %[^n]",l1.l_date);
            gotoXY(10,10);
            printf("New Loan Type: ");
            scanf(" %[^n]",l1.l_type);

```



```

gotoXY(10,12);
    printf("New Loan Amount: ");
    scanf("%d", &l1.l_amt);
gotoXY(10,14);
    printf("New Duration: ");
    scanf("%d", &l1.l_dur);
}
fprintf(temp, "%d|%d|%s|%s|%d|%d\n",l1.l_id, l1.a_no, l1.l_date,l1.l_type, l1.l_amt, l1.l_dur);
}
fclose(fLoan);
fclose(temp);
remove("loan.txt");
rename("temp_loan.txt", "loan.txt");
if (found){
    gotoXY(10,16);
    printf(yellow"Updated!");
}
else{
    gotoXY(10,16);
    printf(red"Not Found!");
}
delay();
return 0;
}

// FILE DELETE OPERATIONS
// Delete data in Account file
int deleteAccount() {
    int found = 0;
    FILE *fAcc = fopen("accounts.txt", "r");
    FILE *temp = fopen("temp_acc.txt", "w");
    int id;
    renderScreen("Delete Account");
    gotoXY(10,4);
    printf(white"Enter Account ID to Delete: ");

```



```

scanf("%d", &id);
while (fscanf(fAcc, "%d|^[^|]|%d\n", &a1.a_id, a1.a_name, a1.a_addr, &a1.a_bal) != EOF) {
    if (a1.a_id == id){
        found = 1;
    }
    if (a1.a_id != id) {
        fprintf(temp, "%d|s|s|s|d\n", a1.a_id, a1.a_name, a1.a_addr, a1.a_bal);
    }
}
if (found == 0){
    printf(red"Cann't find ID...");
    printf(white);
}
fclose(fAcc);
fclose(temp);
remove("accounts.txt");
rename("temp_acc.txt", "accounts.txt");
gotoXY(10,10);
if(found == 1){
    printf(yellow"Deleted!\n");
    printf(white);
}
delay();
return 0;
}
// Delete data in Transaction file
int deleteTransaction() {
    int found = 0;
    FILE *fTran = fopen("transaction.txt", "r");
    FILE *temp = fopen("temp_tran.txt", "w");
    int id;
    renderScreen("Delete Transaction");
    gotoXY(10,4);
    printf(white"Enter Transaction ID to Delete: ");

```



```

scanf("%d", &id);
while (fscanf(fTran, "%d|^[^|]|^[^|]|%d|%\n",&t1.t_id, t1.a_no, t1.t_date, &t1.t_type, &t1.t_amt) !=
EOF) {
    if (t1.t_id == id){
        found = 1;
    }
    if (t1.t_id != id) {
        fprintf(temp, "%d|^[^|]|^[^|]|%d|%\n",
            t1.t_id, t1.a_no, t1.t_date, t1.t_type, t1.t_amt);
    }
}
if (found == 0){
    printf(red"Cann't find ID...");
    printf(white);
}
fclose(fTran);
fclose(temp);
remove("transaction.txt");
rename("temp_tran.txt", "transaction.txt");
gotoXY(10,10);
if(found == 1){
    printf(yellow"Deleted!\n");
    printf(white);
}
delay();
return 0;
}
// Delete data in Loan file
int deleteLoan() {
    int found = 0;
    FILE *fLoan = fopen("loan.txt", "r");
    FILE *temp = fopen("temp_loan.txt", "w");
    int id;
    renderScreen("Delete Loan");

```



```

gotoXY(10,4);
printf(white"Enter Loan ID to Delete: ");
scanf("%d", &id);
while (fscanf(fLoan, "%d|%d|%^]|%[^]|%d|%d\n",&l1.l_id, &l1.a_no, l1.l_date, l1.l_type, &l1.l_amt,
&l1.l_dur) != EOF) {
    if (l1.l_id == id){
        found = 1;
    }
    if (l1.l_id != id) {
        fprintf(temp, "%d|%d|%s|%s|%d|%d\n",
            l1.l_id, l1.a_no, l1.l_date, l1.l_type, l1.l_amt, l1.l_dur);
    }
}
if (found == 0){
    printf(red"Cann't find ID...");
    printf(white);
}
fclose(fLoan);
fclose(temp);
remove("loan.txt");
rename("temp_loan.txt", "loan.txt");
gotoXY(10,10);
if(found == 1){
    printf(yellow"Deleted!\n");
    printf(white);
}
delay();
return 0;
}

// FILE VIEW OPERATIONS
// View data in Account file
int viewAccount() {
    FILE *fAcc = fopen("accounts.txt", "r");
    if (!fAcc) {

```



```

    renderScreen("View Account");
    gotoXY(10,4); printf("No accounts found.");
    getch();
    return 1;
}
page = 1;
while (1) {
    renderScreen("View Account");
    count = 0;
    y = 6;
    gotoXY(10,4);
    printf("Accounts - Page %d", page);
    while (count < 2 && fscanf(fAcc, "%d|%^]|%[^]|%d\n", &a1.a_id, a1.a_name, a1.a_addr, &a1.a_bal)
    != EOF) {
        gotoXY(10, y++); printf("Account ID : %d", a1.a_id);
        gotoXY(10, y++); printf("Name      : %s", a1.a_name);
        gotoXY(10, y++); printf("Address   : %s", a1.a_addr);
        gotoXY(10, y++); printf("Balance   : %d", a1.a_bal);
        gotoXY(10, y++); drawLine('-', 60);
        count++;
        if (y >= 18) break;
    }
    if (count == 0) break;
    nav = bottomBar();
    if (nav == 9) break;
    page++;
}
fclose(fAcc);
return 0;
}

// View data in Transaction file
int viewTransaction() {
    FILE *fTran = fopen("transaction.txt", "r");
    if (!fTran) {

```



```

    renderScreen("View Transaction");
    gotoXY(10,4); printf("No transactions found.");
    getch();
    return 1;
}
page = 1;
while (1) {
    renderScreen("View Transaction");
    count = 0;
    y = 6;
    gotoXY(10,4);
    printf("Transactions - Page %d", page);
    while (count < 2 && fscanf(fTran, "%d|%^]|%[^]|%d|%%d\n",&t1.t_id, t1.a_no, t1.t_date, &t1.t_type,
&t1.t_amt) != EOF) {
        gotoXY(10, y++); printf("Transaction ID : %d", t1.t_id);
        gotoXY(10, y++); printf("Account No   : %s", t1.a_no);
        gotoXY(10, y++); printf("Date       : %s", t1.t_date);
        gotoXY(10, y++); printf("Type       : %d", t1.t_type);
        gotoXY(10, y++); printf("Amount    : %d", t1.t_amt);
        gotoXY(10, y++); drawLine('-', 60);
        count++;
        if (y >= 18) break;
    }
    if (count == 0) break;
    nav = bottomBar();
    if (nav == 9) break;

    page++;
}
fclose(fTran);
return 0;
}
// View data in Loan file
int viewLoan() {

```



```

FILE *fLoan = fopen("loan.txt", "r");
if (!fLoan) {
    renderScreen("View Loan");
    gotoXY(10,4); printf("No loans found.");
    getch();
    return 1;
}
page = 1;
while (1) {
    renderScreen("View Loan");
    count = 0;
    y = 6;
    gotoXY(10,4);
    printf("Loan Records - Page %d", page);
    while (count < 2 && fscanf(fLoan, "%d|%d|^[^|]|^[^|]|%d|%d\n", &l1.l_id, &l1.a_no, l1.l_date,
l1.l_type,&l1.l_amt, &l1.l_dur) != EOF) {
        gotoXY(10, y++); printf("Loan ID    : %d", l1.l_id);
        gotoXY(10, y++); printf("Account No  : %d", l1.a_no);
        gotoXY(10, y++); printf("Date       : %s", l1.l_date);
        gotoXY(10, y++); printf("Type       : %s", l1.l_type);
        gotoXY(10, y++); printf("Amount     : %d", l1.l_amt);
        gotoXY(10, y++); printf("Duration   : %d months", l1.l_dur);
        gotoXY(10, y++); drawLine('-', 60);
        count++;
        if (y >= 18) break;
    }
    if (count == 0) break;
    nav = bottomBar();
    if (nav == 9) break;
    page++;
}
fclose(fLoan);
return 0;
}

```